

Parv Patel

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Education

Indian Institute of Technology Palakkad

Bachelor of Technology in Data Science

CGPA: 8.34/10

Palakkad, Kerala

Jul 2023 – Jun 2027

Experience

IIT Palakkad — Project Intern

Apr 2026 – Present

AI–Infant Health & Wellness Sense, funded by IIT Indore Drishti CPS Foundation under DISHA 2.0

- Developed a React Native clinician dashboard using a decoupled ref-buffering pipeline to render high-fidelity PPG waveforms at 60 FPS, integrating on-device DSP filters and peak detection while offloading 5s pulse blocks to a FastAPI model for blood pressure classification.

Accenture — Advanced Engineering Hub Intern

May 2026 – July 2026

Bengaluru

- Contributed to the development of an enterprise Identity & Access Management (IAM) solution by implementing explainable machine learning pipelines for identity risk scoring and access governance.

Projects

DualForensics: Deepfake Detection System | Live Demo | 🐙 GitHub | *PyTorch, FastAPI, Docker, OpenCV*

- Architected a containerized REST inference microservice (FastAPI + Docker) wrapping a CNN-Transformer deepfake classifier, handling video upload through preprocessing, inference, and explainability rendering.
- Built a standalone OpenCV preprocessing pipeline (frame sampling, face detection, crop normalization) decoupled from the model layer, with model-artifact loading separated from the service lifecycle via a dedicated `/api/health` check for independent deployability.
- Enforced production-grade reliability: strict payload/format validation, exception-safe temp-file cleanup, graceful model-load failure handling (clean HTTP 503s), and single-threaded OMP/MKL/PyTorch tuning to eliminate CPU contention under concurrent requests.
- Achieved 0.9787 AUC on FaceForensics++ and 83% accuracy on unseen samples; compressed 16-frame Grad-CAM explainability output into a single optimized base64 payload to minimize response size.

Financial Sentiment Forecasting using Entity-Aware Transformers | 🐙 GitHub | *Python, PyTorch, Transformers, XGBoost, LightGBM, Scikit-Learn*

- Built an end-to-end ML pipeline over 25K+ entity-level financial news instances, combining entity-aware sentiment extraction, temporal feature engineering, and leakage-safe return forecasting across 1/2/5-day horizons.
- Fine-tuned DeBERTa-v3 for aspect-based sentiment classification (92%+ accuracy), producing confidence, entropy, and embedding features consumed by downstream forecasting models.
- Decoupled the pipeline into a feature-extraction stage (DeBERTa outputs) and a modeling stage, enabling the same feature set to be swapped across 6+ downstream models without retraining the transformer.
- Designed leakage-safe temporal aggregation using causally shifted rolling windows; benchmarked 6+ models (Ridge, XGBoost, LightGBM, PyTorch MLP) with ablation studies and 2,000-sample bootstrap significance testing.

Technical Skills

Languages: Python, Java, SQL, C++, JavaScript/TypeScript

Libraries & Tools: PyTorch, Scikit-Learn, Pandas, NumPy, Git, Jupyter

Machine Learning & Deep Learning: Regression, Classification, Clustering, PCA, Ensemble Methods, CNNs, Transformers, Attention Mechanisms, BERT, Feature Engineering, Hyperparameter Optimisation, Statistical Inference

Backend & Systems: FastAPI, Docker, REST API Design, OpenCV

Data Structures & Algorithms: Arrays, Linked Lists, Trees, Graphs, Stack&Queue, Dynamic Programming

CS Fundamentals: Operating Systems, Database Management Systems, Computer Networks

Achievements

4th Rank, Inter IIT Tech Meet 14.0 – Built a GAN-based synthetic tabular data generator with topology-preserving losses, competing among 20+ teams across IITs (Genuity IO).

Positions of Responsibility

Convener, The Integral Cup

Nov 2025 – May 2026

Directed operations and logistics for a national academic competition spanning 26+ centres (IITs, BITS, IIITs, IISc).